Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: Reflected image always travels in the opposite direction of object.

Reason: Speed of the point image formed by reflected rays is always equal to the speed of its point object.

A. both assertion and reason are true but reason is not correct explanation of assertion.

B. assertion is true, but reason is false.

C. assertion is false, but reason is true.

D. both assertion and reason are false.

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the path difference between two waves at a point is 29λ, then the point will be,

A. bright

B. dark

C. neither bright nor dark.

D. invisible

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A man who wears glasses of power 3 diopters must hold a newspaper at least 25 cm away to see the print clearly. How far away would the newspaper have to be if he took off the glasses and still wanted clear vision?

A. Hold the paper 1m away

B. Hold the paper 33.3cm away

C. Hold the paper 100m away

D. Hold the paper 2m away

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The light waves from two independent monochromatic light sources are given by, $y_1 = 2\sin\omega t$ and $y_2 = 3\cos\omega t$. Then, which is the correct statement?

A. Both the waves are coherent.

B. Both the waves are incoherent.

C. Both the waves are in the same phase.

D. Both the waves have different time period.

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When white light passes through a dispersive medium, it breaks up into various colours. Which of the following statements is true?

A. Velocity of light for violet is greater than the velocity of light for red colour.

B. Velocity of light for violet is less than the velocity of light for red.

C. Velocity of light is the same for all colours

D. Velocity of light for different colours has nothing to do with the phenomenon of dispersion

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : When a tiny circular obstacle is placed in the path of light from some distance, a bright spot is seen at the centre of shadow of the obstacle.
Reason : Destructive interference occurs at the centre of the shadow.

A. both assertion and reason are true and the reason is the correct explanation of the assertion.

B. both assertion and reason are true but reason is not the correct explanation of the assertion

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assertion is true but
C. reason is false.

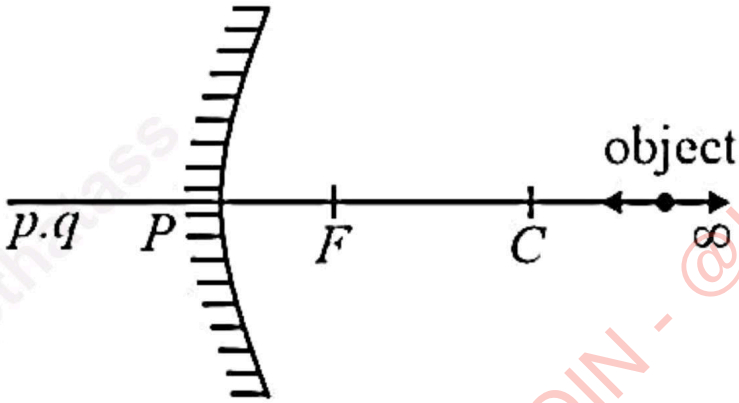
the assertion and
D. reason both are false.

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A point object is moving along the principal axis of a concave mirror with uniform velocity towards the pole. Initially, the object is at infinite distance from pole on right side of the mirror as shown. Before the object collides with the mirror, how many times the distance between the object and its image is 40 cm?



A. One time.

B. Two times.

C. Three times

D. Data insufficient.

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Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a YDSE, if the slits are of unequal widths,

A. fringes will not be formed

B. the positions of minimum intensity will not be completely dark

C. bright fringe will not be formed at the centre of the screen

D. distance between two consecutive bright fringes will not be equal to the distance between two consecutive dark fringes.

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A thin prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of the prism P_2 is:

A. 5.33°

B. 4°

C. 3°

D. 2.6°

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Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two beams, A and B of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity) a rotation of polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals:

A. $\frac{1}{3}$

B. 3

C. $\frac{3}{2}$

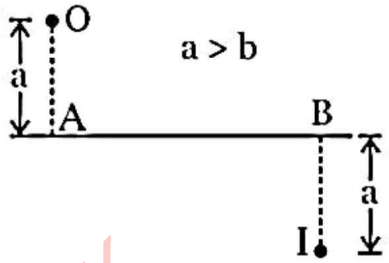
D. 1

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A luminous point is placed at O, whose image is formed at I as shown in diagram. Line AB is optic axis and its length is given. Which of the following statements are correct?



A. If the lens used to obtain the image, the lens must be a converging lens and position of its principal focus can be found out

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If the mirror is used to obtain the image, the mirror must be a convex mirror whose pole will be at point of intersection of line OI

B. with the optic axis

Position of principal focus of the mirror

C. cannot be found out

I is a real image of

D. virtual object

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In YDSE, the slits have different widths. As a result, the amplitude of waves from two slits is A and $2A$ respectively. If I_0 be the maximum intensity of the interference pattern then the intensity of the pattern at a point where the phase difference between waves is ϕ is given by $\frac{I_0}{P}\left(5+4\cos\phi\right)$. Where P is an integer. Find the value of P?

A. 9

B. 3

C. 6

D. 1

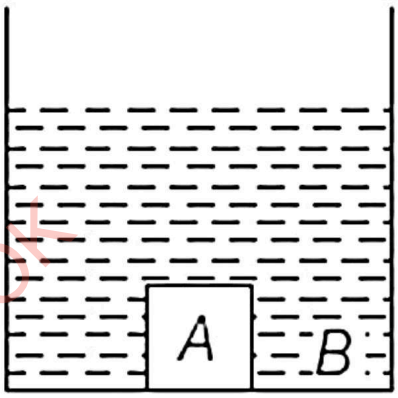
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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A parallel sided block of glass of refractive index 1.5, which is 36 mm thick rests on the bottom of a tank, filled with water (refractive index $= 4/3$). The difference between the apparent depth of bottom at A and B, when seen from vertically above is equal to



A. 2 mm

B. 3 mm

C. 4 mm

D. 6 mm

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The focal length of objective of a telescope is 50 cm and eye-piece is 5 cm. The telescope is focused from a distinct vision on a scale 2 m away. The separation between the objective and the eye-piece is, if final image is formed at the least distance of distinct vision,

A. 75cm

B. 60cm

C. 71cm

D. 74cm

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Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In Fraunhofer diffraction pattern, slit width is 0.2 mm and screen is at 2 m away from the lens. If wavelength of light used is 5000Å then the distance between the first minimum on either side the central maximum is (θ is small and measured in radian)

A. 10^{-1} m

B. 10^{-2} m

C. 2×10^{-2} m

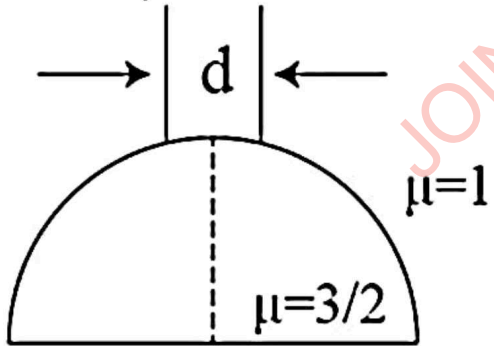
D. 2×10^{-1} m

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A beam of diameter d is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to d, then the diameter of the beam at the base of the hemisphere will be



A. $\frac{3}{4}d$

B. d

C. $\frac{d}{3}$

D. $\frac{2}{3}d$

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A polarizer - analyser set is adjusted such that the intensity of light coming out of the analyser is just 36% of the original intensity. Assuming that the polarizer - analyser set does not absorb any light, the angle by which the analyser needs to be rotated further, to reduce the output intensity to zero, is $\left(\sin^{-1}\left(\frac{3}{5}\right) = 37^\circ\right)$

A. 53°

B. 37°

C. 90°

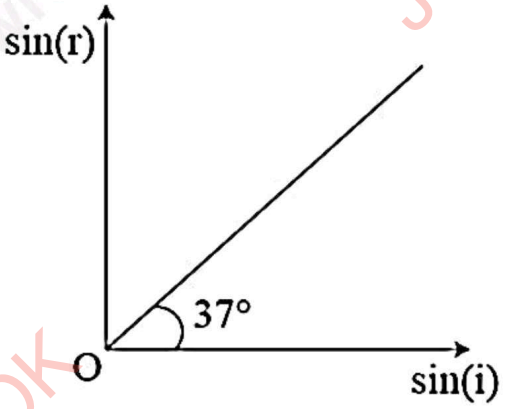
D. 45°

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A ray of light, travelling in medium A, is incident on plane interface of two media A and B and gets refracted into medium B. The angle of incidence is i and that of refraction is r. Graph between $\sin(i)$ and $\sin(r)$ is as shown in diagram. The correct statement among the following is



Speed of light in medium B is three fourth of that in

A. medium A

Total internal reflection can take place

B.

Refraction index of medium A is greater than that of medium B

C.

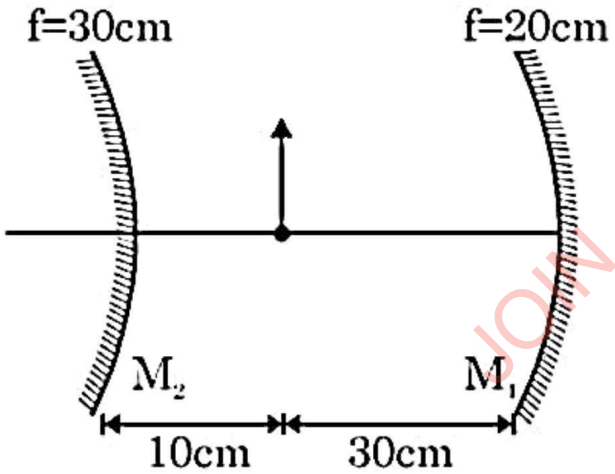
D. None of these

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the figure shown find the total magnification after two successive reflections first on M_1 and then on M_2. (Assume paraxial rays only)



A. $+6$

B. -6

C. $+3$

D. -3

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: A spherical wavefront is produced by a point source of light.

Reason: It is because, the locus of all points, which are equidistant from the point source, is a sphere.

A. Assertion is true, reason is true; reason is a correct explanation for assertion.

B. Assertion is true, reason is true; reason is not a correct explanation for assertion.

C. Assertion is true, reason is false.

D. Assertion is false, reason is true.

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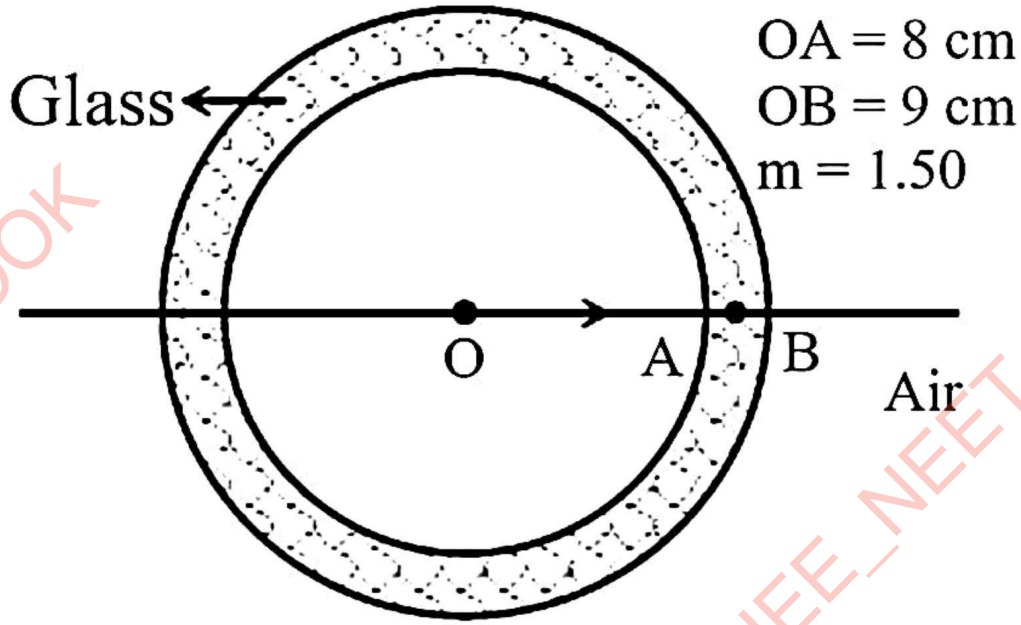
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

A small filament is at the centre of a hollow glass sphere of inner and outer radii 8 cm and 9 cm respectively. The refractive index of glass is 1.50. Calculate the position of the image of the filament when viewed from outside the sphere.



Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

If one face of a prism angle 30 and $\mu = \sqrt{2}$ is silvered, the incident ray retraces its initial path. What is the angle of incidence?

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

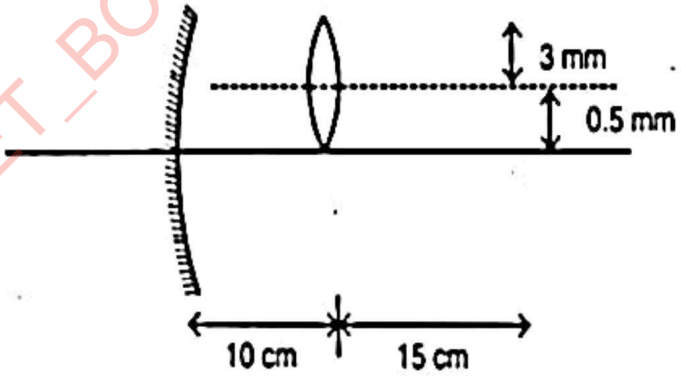
A beam of light consisting of two wavelengths 450 nm and 750 nm is used to obtain interference fringes in Young's double slit experiment. The separation between the slits is 1.0 mm and the distance between the plane of the slits and the screen is 100 cm. The least distance from the central maximum where the bright fringes due to both the wavelengths coincide is $w \times 10^{-5}$ metre. find w

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A concave mirror of focal length 20 cm and a convex lens of focal length 10 cm are kept with their optic axes parallel but separated by 0.5 cm as shown in the figure. The distance between the lens and mirror is 10 cm. An object of height 3 mm is placed on the optic axis of lens at a distance 15 cm from the lens. Find the length of the image formed by the mirror in mm.



Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

In Young's double-slit experiment, the separation between the slits is $d = 0.25$ cm and the distance of the screen from the slits is $D = 100$ cm. When the wavelength of light used in the experiment is $\lambda = 6000$Å, the intensity at a distance $x = 4 \times 10^{-5}$ m from the central maximum is $\frac{p}{q} I_0$, where I_0 is the intensity of central maxima and p and q are the smallest integers. What is the value of $p + q$?

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Sucrose is a disaccharide and a non-reducing sugar.
Reason (R): Sucrose involves glycosidic linkage between C_1 of β-glucose and C_2 of α-fructose.
Choose the most appropriate answer from the options given below :

A. Both (A) and (R) are true and (R) is the true explanation of (A).

B. (A) is true but (R) is false.

C. (A) is false but (R) is true.

D. Both (A) and (R) are true but (R) is not the true explanation of (A).

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The isomer(s) of $\left[Co(NH_3)_4Cl_2\right]$ that has/ have a $Cl - Co - Cl$ angle of 90°, is/are:

A. meridional and trans

B. cis and trans

C. trans only

D. cis only

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_2.6H_2O(X)$ and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as $1:3$ electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y. Among the following options, which statement is incorrect?

A. The hybridization of the central metal ion in Y is d^2sp^3

B. Addition of silver nitrate to Y gives only two equivalents of silver chloride

C. When X and Z are in equilibrium at $0^\circ C$, the colour of the solution is pink

D. Z is a tetrahedral complex

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The hormone that helps in the conversion of glucose to glycogen is

A. cortisone

B. bile acids

C. adrenaline

D. insulin

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Statement-1: The geometrical isomers of complex $[M(NH_3)_4Cl_2]$ are optically inactive

Statement-2: Both geometrical isomers of the complex $[M(NH_3)_4Cl_2]$ possess axis of symmetry.

A. If both statement-1 and statement-2 are true and statement-2 is a correct explanation of statement-1.

B. If both statement-1 and statement-2 are true and statement-2 is **not** a correct explanation of statement-1.

C. Statement-1 is true and statement-2 is false.

D. Statement-1 is false and statement-2 is true.

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$C_1 - C_4$ glycosidic bond is NOT found in

A. Sucrose

B. Lactose

C. Starch

D. Maltose

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The correct order of intensity of colors of the compounds is:

A. $\left[\mathrm{Ni(CN)_4}\right]^{2-} > \left[\mathrm{NiCl_4}\right]^{2-} > \left[\mathrm{Ni(H_2O)_6}\right]^{2+}$

B. $\left[\mathrm{Ni(H_2O)_6}\right]^{2+} > \left[\mathrm{NiCl_4}\right]^{2-} > \left[\mathrm{Ni(CN)_4}\right]^{2-}$

C. $\left[\mathrm{NiCl_4}\right]^{2-} > \left[\mathrm{Ni(H_2O)_6}\right]^{2+} > \left[\mathrm{Ni(CN)_4}\right]^{2-}$

D. $\left[\mathrm{NiCl_4}\right]^{2-} > \left[\mathrm{Ni(CN)_4}\right]^{2-} > \left[\mathrm{Ni(H_2O)_6}\right]^{2+}$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion : $\left[Ti(\mathrm{H_2O})_6\right]^{3+}$ is coloured while $\left[\mathrm{Sc(H_2O)_6}\right]^{3+}$ is colourless.

Reason : $d-d$ transition is not possible in $\left[\mathrm{Sc(H_2O)_6}\right]^{3+}$.

A. If both assertion and reason are true and the reason is the correct explanation of the assertion.

B. If both assertion and reason are true but reason is not the correct explanation of the assertion.

C. If assertion is true but reason is false.

D. If the assertion and reason both are false.

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements is not correct?

A. Proteins are polyamides formed from amino acids

B. Except glycine, all other amino acids show optical activity

C. Natural proteins are commonly made up of L-isomer of amino acids

D. In α -amino acids, $-NH_2$ and $-COOH$ groups are attached to different carbon atoms

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The IUPAC name of the following complex is $\left[Cr(NH_3)_3(H_2O)_2Cl\right]Cl_2$.

A. Triamminediaqua chlorido chromium (III) chloride.

B. Diaquatriammine chlorido chromium (III) chloride.

C. Chlorido diaquatriammine chromium (III) chloride.

D. Triammine diaqua trichlorido chromium (III).

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the characteristics is not common between $\left[Cu(en)_2\right]^{2+}$ and $\left[Ni(dmg)_2\right]$?

A. Geometry of complexes

B. Hybridisation of central metal cation

C. Magnetic behaviour

D. Number of stereoisomers

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A nucleoside on hydrolysis gives

A. an aldopentose and a nitrogenous base

B. an aldopentose and phosphoric acid

C. an aldopentose, a nitrogenous base and phosphoric acid

D. a nitrogenous base and phosphoric acid

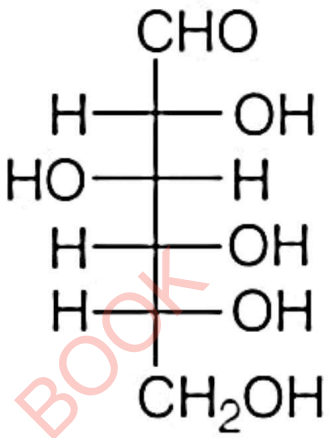
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Question No. 13

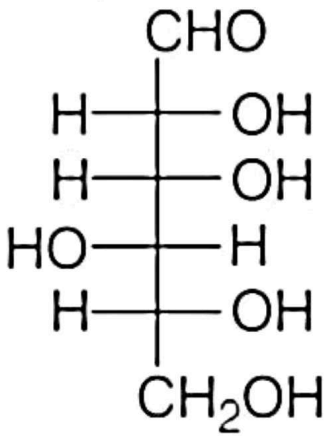
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

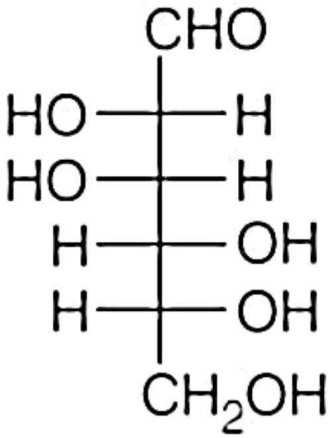
Consider the following carbohydrate molecules:



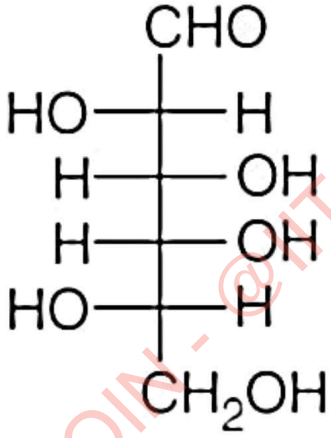
(A)



(B)



(C)



(D)

Among the given molecules, the epimeric pair is

A. A and B

B. A and D

C. C and B

D. C and A

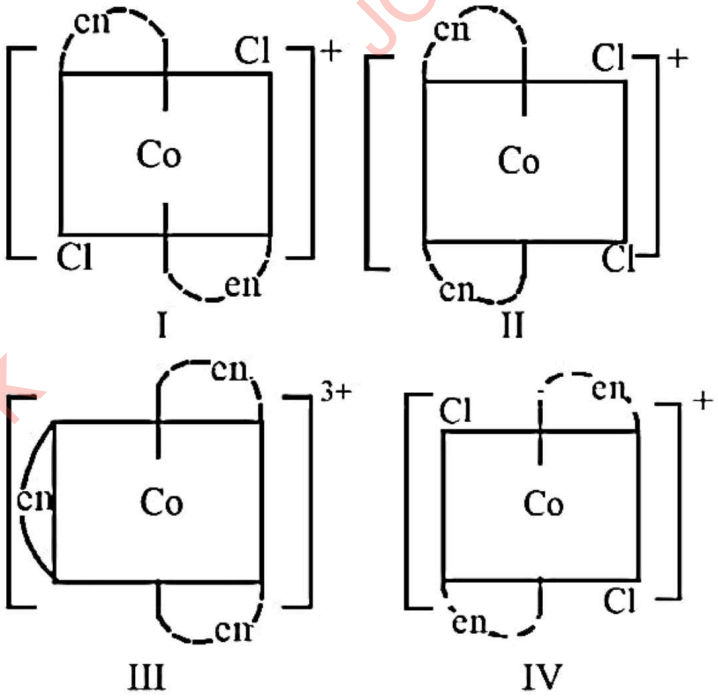
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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following ions are optically active?



A. I only

B. II only

C. II and III

D. IV only

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The pyrimidne bases present in DNA are

A. cytosine and adenine

B. cytosine and guanine

C. cytosine and thymine

D. cytosine and uracil

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Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which one of the following complexes is an outer orbital complex

A. $\left[\mathrm{Co}(\mathrm{NH}_3)_6\right]^{3+}$

B. $\left[Mn(CN)_6\right]^{4-}$

C. $\left[Fe(CN)_6\right]^{4-}$

D. $\left[Ni(NH_3)_6\right]^{2+}$

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The deficiency of vitamin B_1 causes

A. Beri-beri

B. Scurvy

C. Rickets

D. Anaemia

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following organometallic compound is σ and π bonded

A. $Fe(CH_3)_3$

B. $[Co(CO)_5NH_3]^{2+}$

C. $Pb(CH_3)_4$

D. $Al(CH_3)_3$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The bond in $K_4[Fe(CN)_6]$ are:

A. All ionic

B. All covalent

C. Ionic and covalent

D. Ionic, covalent and co-ordiante covalent

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A compound of molecular weight 180 is acetylated to give a compound of molecular weight 390. The number of amino groups in the initial compound is

A. 2

B. 4

C. 5

D. 6

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Among the complex ions, $[Co(NH_2 - CH_2 - CH_2 - NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, [Fe
and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) cis-trans isomerism is

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$\left[Fe(H_2O)_4(OH)_2\right]^+$, $\left[Fe(NH_3)_2(CN)_4\right]^-$, $\left[Co(NH_2-CH_2-CH_2$

:

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$^{+}$, $[Fe(NH_3)_2(CN)_4]^-$, $[Co(NH_2-CH_2-CH_2-NH_2)_2(NH_3)Cl]^{2+}$

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Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

The open-chain glucose, (an aldohexose) and fructose (a $2-$ ketohexose) have X and Y chiral carbons respectively. Find the sum of $\mathrm{X}+\mathrm{Y}$.

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

A weak field complex of Ni^{2+} has magnetic moment of 2.82BM. The number of electron in the $\mathrm{t}_{2\,\mathrm{g}}$ level of Ni^{2+} will be

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Among the following, the number of essential amino acids is_____.

Arginine, threonine, valine, proline, methionine, lysine, tryptophan, serine

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

How many statements is/are correct for given complex $\left[\mathrm{Co}(\mathrm{C}_2\mathrm{O}_4)_3\right]^{3-}$

i) It contain negative ligands only
ii) The oxidation state of central metal atom is $+3$
iii) It is optically active
iv) Due to chelation the stability of the complex increases
v) It exhibit geometrical isomerism
vi) It is octahedral with $\mathrm{d}^2\mathrm{sp}^3$ hybridization.
vii) All oxalate ligands are replaced by ethylenediamine, then complex do not show optical activity.

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Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solution of the differential equation $\frac{dy}{dx} + x(2x+y) = x^3(2x+y)^3 - 2$ is (C being an arbitrary constant)

A. $\frac{1}{2x+xy} = x^2 + 1 + Ce^x$

B. $\frac{1}{(2x+y)^2} = x^2 + 1 + Ce^{x^2}$

C. $\frac{1}{2x+y} = x + 1 + Ce^{-x^2}$

D. $\frac{1}{(2x+y)^2} = x^2 + 1 + C$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The area (in sq. units) of the region $\{x \in R : x \geq 0, y \geq 0, y \geq x - 2 \text{ and } y \leq \sqrt{x}\}$ is

A. $\frac{13}{3}$

B. $\frac{8}{3}$

C. $\frac{10}{3}$

D. $\frac{5}{3}$

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solution of the differential equation $\sin y e^x dx - e^x \cos y dy = \sin^2 y dx$ is (where, c is an arbitrary constant)

A. $e^x \sin y = x + c$

B. $e^x = (x + c) \sin y$

C. $e^x \cdot x = \sin y + c$

D. $e^x \cdot \sin y = x^2 + c$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The area (in sq. units) bounded by $[|x|] + [|y|] = 2$ in the first and third quadrant is (where, $[.]$ is the greatest integer function),

A. 4

B. 3

C. 6

D. 10

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find the area bounded by the curve $y = xe^{-x}$, $xy = 0$ & $x = c$ where c is the x-coordinate of the curve where second derivative vanishes.

A. $1 - 3e^{-2}$ sq. units

B. $1 - 3e^{-1}$ sq. units

C. $1 - 4e^{-2}$ sq. units

D. $1 - 4e^{-1}$ sq. units

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The order of the differential equation of the family of curves $y = \frac{a}{c}\sin(bx) + 3^{dx}$ where a, b, c, d are arbitrary constants is

A. 4

B. 3

C. 2

D. 1

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Solution of the differential equation $\frac{x+\frac{x^3}{3!}+\frac{x^5}{5!}\cdots}{1+\frac{x^2}{2!}+\frac{x^4}{4!}\cdots}=\frac{dx-dy}{dx+dy}$ is

A. $2ye^{2x}=ce^{2x}+1$

B. $2ye^{2x}=2ce^{2x}-1$

C. $ye^{2x}=ce^{2x}+2$

D. None of these

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The area (in square units) enclosed by $|y|-|x|=1$ and $x^2+y^2=1$ is

A. 2

B. zero

C. infinite

D. None of these

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Solution of the equation $x^2y - x^3\frac{dy}{dx} = y^4\cos x$, when $y(0) = 1$ is

A. $y^3 = 3x^3\sin x$

B. $x^3 = 3y^3\sin x$

C. $x^3 = y^3\sin x$

D. $x^3 = y^3\cos x$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solution of the differential equation $xy^2\left(y_1^2 + 2\right) = 2y_1y^3 + x^3$ is

A. $\left(x + y - a\right)\left(x^2 - y^2 - bx^2\right) = 0$

B. $\left(x^2 - y^2 - a\right)\left(x^2 - y^2 + bx^4\right) = 0$

C. $\left(x^2 + y^2 - a\right)\left(x^2 + y^2 - bx^4\right) = 0$

D. none of these

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Area bounded by $y = -x^2 + 6x - 5, y = -x^2 + 4x - 3$ and $y = 3x - 15$, for $x > 1$, is -

A. 73

B. $13/6$

C. $73/6$

D. none of these

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be a differentiable function satisfying the equation $\frac{f'(x)}{2} = \frac{x}{e^{f(x)}}\ \forall x \in R$. If $f'(1) = 1$, then the number of solution(s) of the equation $f'(x) = f(x)$ is

A. 1

B. 0

C. 2

D. 3

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The area bounded by the x-axis, part of the curve $y = 1 + x^{-2}$ and the ordinates $x = 1$ & $x = 2$ is divided into equal parts by the ordinate at $x = a$ such that $a = \frac{3+\sqrt{p}}{8}$, then the value of p is -

A. 70

B. 71

C. 72

D. 73

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the area bounded by f(x) = max(sin x, cos x);
$0 \leq x \leq \frac{\pi}{2}$, $x = \frac{\pi}{2}$ and the coordinates axes is equal to k, then [k + 3] is equal to (where, [.] denotes the greatest integer function

A. 2

B. 8

C. 4

D. 6

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Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider a function $f(x) = \int_0^x [\cos t]dt, x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ then area bounded by curve $y = f(x)$, x axis and lines $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ is (where [.] denotes greatest integer function) :-

A. π

B. π^2

C. $\frac{\pi^2}{2} - 1$

D. $\frac{\pi^2}{2}$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$ is a parameter is of order and degree as follows,s

A. order 2, degree 2

B. order 1, degree 3

C. order 1, degree 1

D. order 1, degree 2

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The area of the smaller portion above x-axis bounded by the curves $y^2 = 8x$ and $\frac{(x-2)^2}{4} + \frac{y^2}{16} = 1$, is -

A. $4\pi - \frac{32}{3}$

B. $8\pi - \frac{16}{3}$

C. $8\pi - \frac{64}{3}$

D. $2\pi - \frac{16}{3}$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The equation of the curve passing through the origin, in the form $y = f(x)$, which satisfies the differential equation $\frac{dy}{dx} = \sin(10x + 6y)$, is

A. $y = \frac{1}{3}\left\{\tan^{-1}\frac{5\tan 4x}{4-3\tan 4x} - 5x\right\}$

B. $y = \frac{1}{2}\left\{\tan^{-1}\frac{4\tan 2x}{2+3\tan 4x} - 4x\right\}$

C. $y = \frac{2}{3}\left\{\tan^{-2}\frac{8\tan 4x}{4+3\tan 4x} + 5x\right\}$

D. $y = \frac{1}{4}\left\{\tan^{-1}\frac{2\tan 5x}{4-2\tan 4x} + 2x\right\}$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solution of the differential equation $(1-x^2)\cdot\frac{dy}{dx}+xy=(x-x^3)y^{\frac{1}{2}}, \forall(|x|<1)$ is $\sqrt{9y}=-f(x)+c(1-x^2)^{\frac{1}{4}}$, where c is an arbitrary constant and $f\left(\frac{1}{2}\right)=\frac{3}{4}$. Then, $f(x)$ is

A. an odd function

B. an even function

C. a periodic function

D. symmetric about line $x=1$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The solution of $\frac{dy}{dx}+\frac{1}{x}=\frac{e^y}{x^2}$ iss

A. $2x=\left(1+Cx^2\right)e^y$

B. $x=\left(1+Cx^2\right)e^y$

C. $2x^2=\left(1+Cx^2\right)e^{-y}$

D. $x^2=\left(1+Cx^2\right)e^{-y}$

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Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

If the area bounded by x-axis, y-axis, $x = 2, y = 2$ & $xy \leq 2 \leq 8(x^2 + y^2)$ is $\alpha - \frac{\pi}{\beta} + \ell n\gamma$ where $\alpha, \beta, \gamma \in \mathbb{N}$ then $\beta - 2(\alpha + \gamma)$ is equal to

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

If $(2xy - y^2 - y)dx = (2xy + x - x^2)dy$ and $y(1) = 1$, then the value of $12|y(-1)|$ is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

The degree of differential equation satisfying the relation $\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda\left(x\sqrt{1+y^2} - y\sqrt{1+x^2}\right)$ is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

The degree of differential equation satisfying the relation $\sqrt{1+x^2}+\sqrt{1+y^2}=\lambda\left(x\sqrt{1+y^2}-y\sqrt{1+x^2}\right)$ is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

The slope at any point of a curve $y=f(x)$ is given by $\frac{dy}{dx}=3x^2$ and it passes through $(-1,1)$. The equation of the curve is $y=x^n+2$ then find n.

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

If the area bounded by $y=\left||x|^2-4|x+3\right|$ and the x-axis from $x=1$ to $x=3$ is $\frac{p}{q}$ (where, p & q are coprime), then the value of $p+q$ is

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